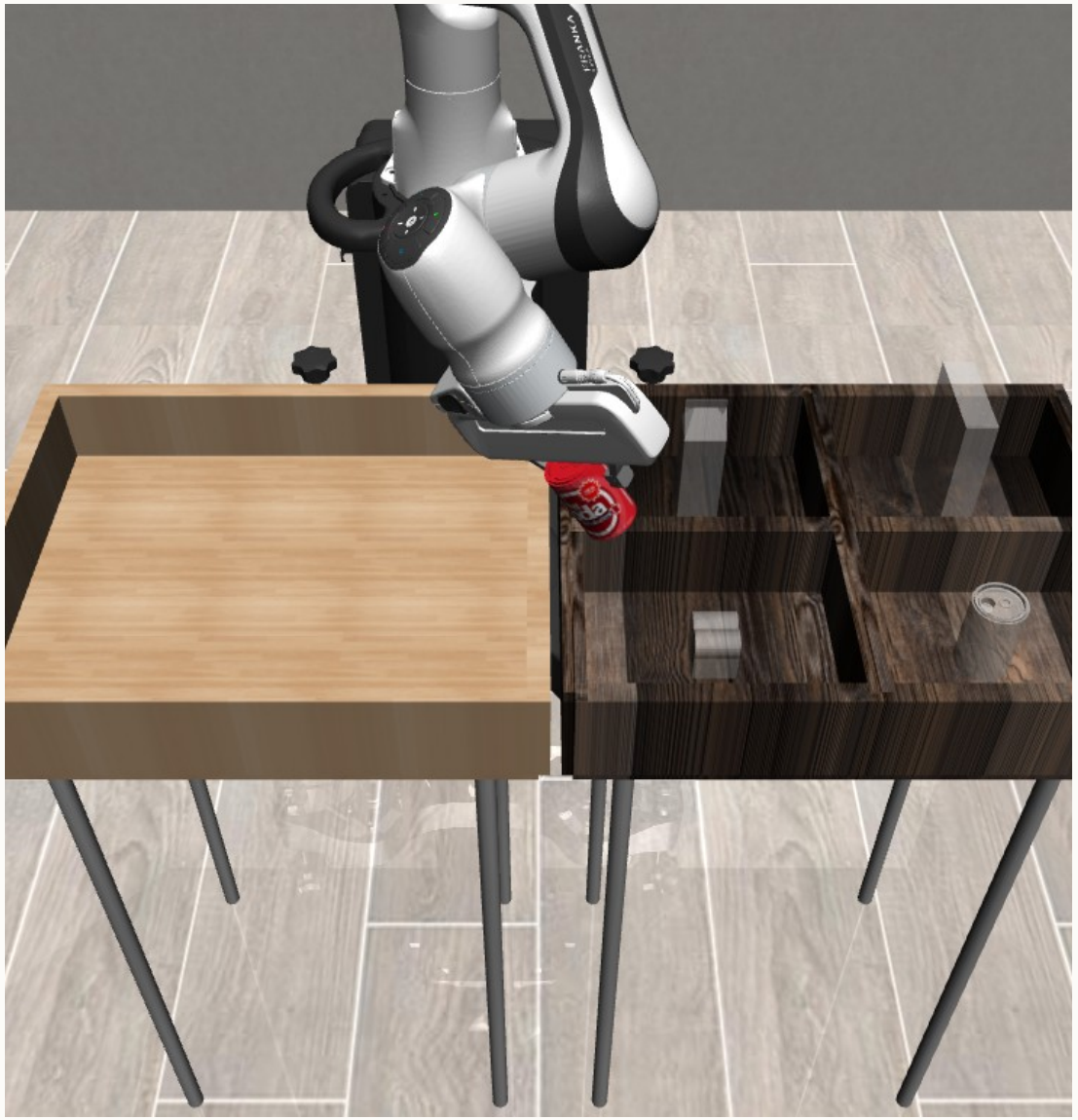
Introduction to Robotics

Manipulation Learning Lab

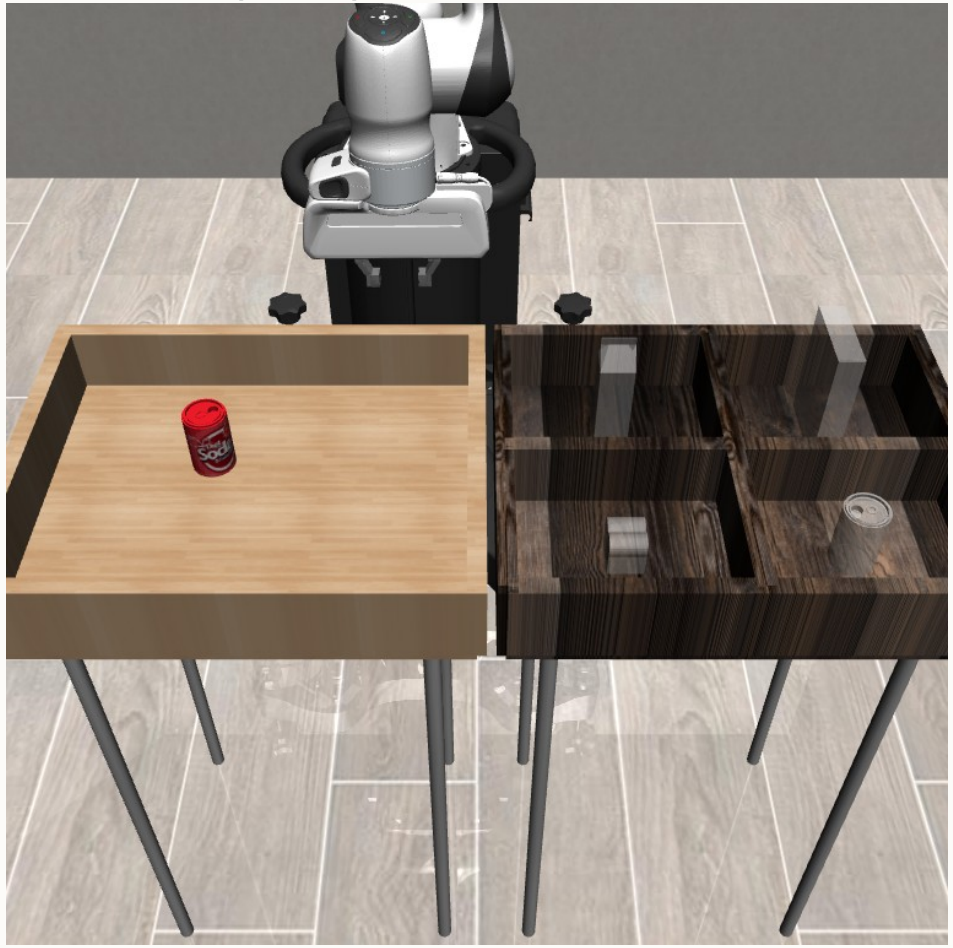
From Demonstrations to Robot Policies



Learning Robot Manipulation from Demonstrations

Can it put the can away?

Panda arm: pick up the can and release it in the marked bin.



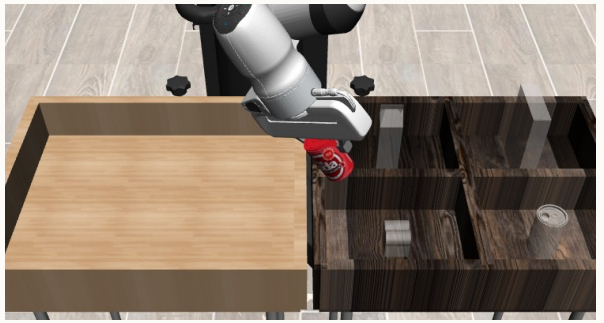
Video in PPTX · 15 s · half speed



Reach



Grasp



Transport



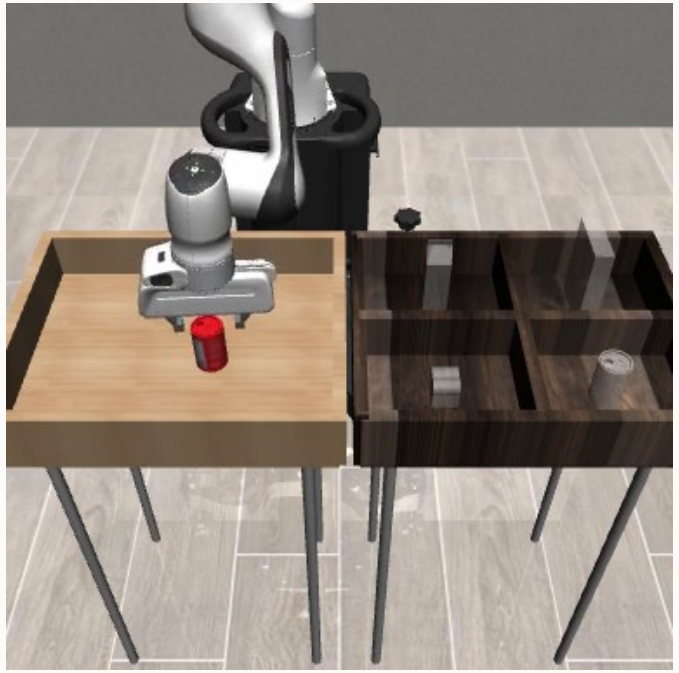
Place

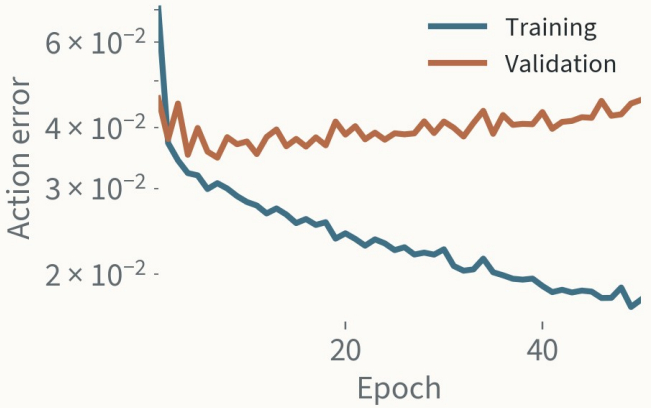
In the target. Released. Stable for 1 second.

Open gripper · no contact · displacement ≤1 cm

How does it learn?

A demonstration pairs the current observation with a human's next action.







Watch a person

Demonstration · Video in PPTX

Fit a policy

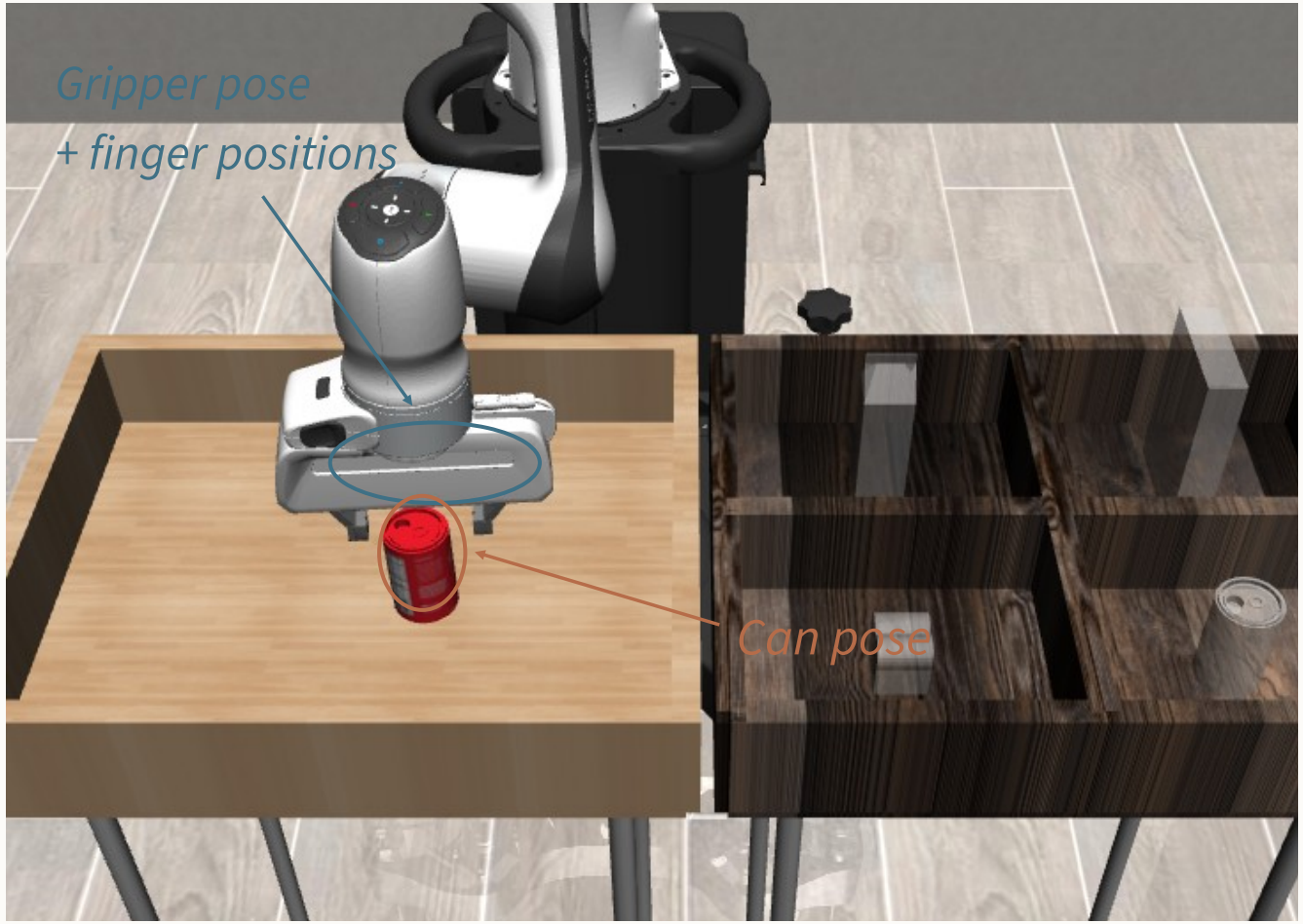
Training

160 train / 20 validation / 20 test demonstrations

Let the robot try

Inference · 20 actions / second

What can the policy see?



A fresh observation, 20 times per second.

16

state values

Gripper pose (7) + finger joints (2)
+ can pose (7)

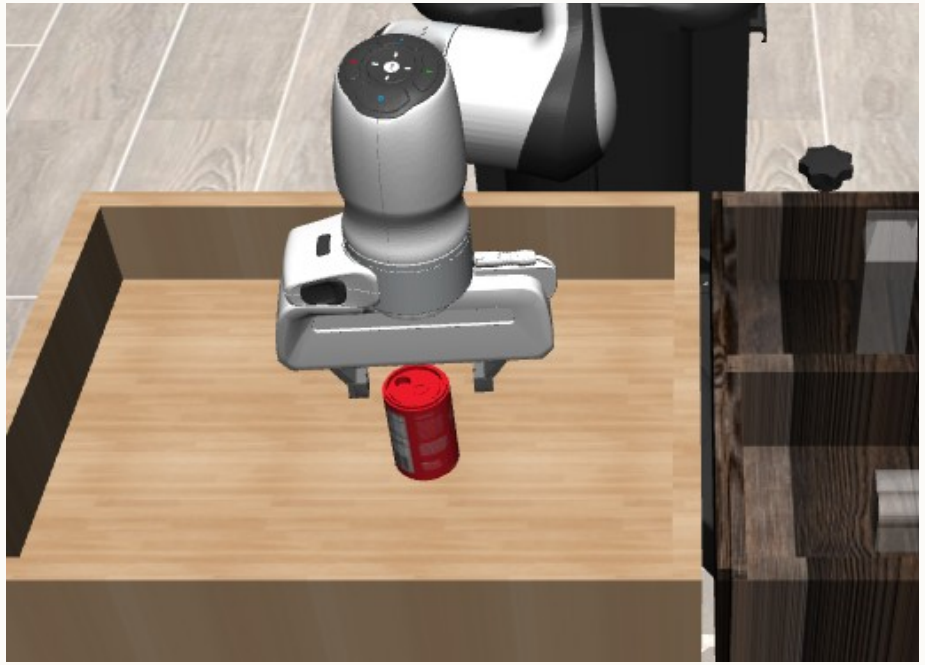
7

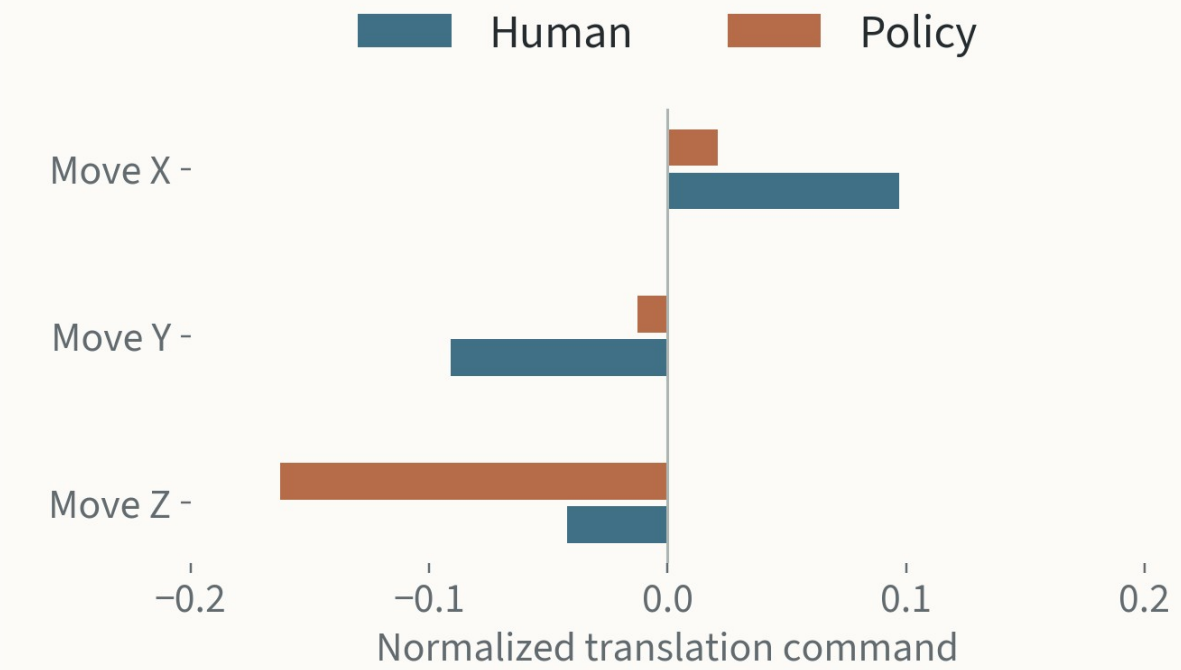
commands

3 translation + 3 rotation + 1 gripper
Normalized commands in [−1, 1]

Learn to match the next move

Behavior cloning trains a neural network to predict the human's next action.





The same observation

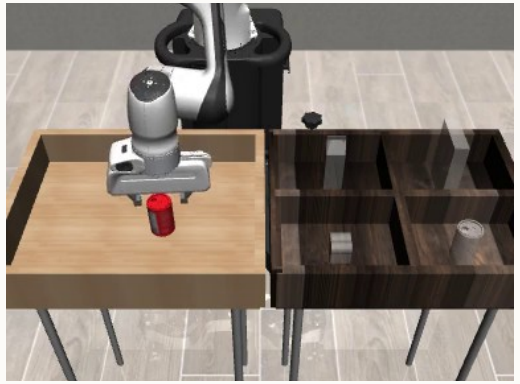
Three hidden layers × 256 units

Adam updates weights; validation selects the model.

$$\mathcal{L} = \text{mean}[(a_{\text{robot}} - a_{\text{human}})^2]$$

Average squared error across samples and all 7 action values.

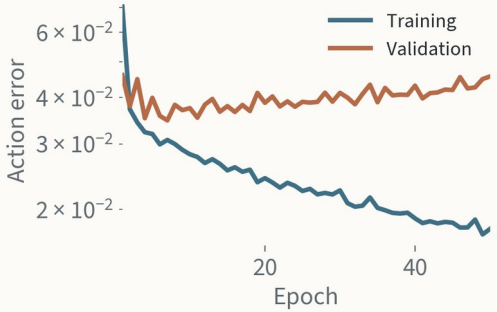
Four things to try



01

Watch

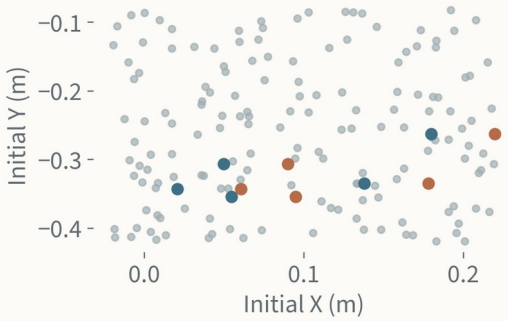
Inspect observations and actions. Q1



02

Train twice

Save both models and curves. Q2–Q3



03

Compare

Use the same test cases. Q4



04

Explain

Inspect at least 3 steps. Q5

Change only the learning rate: 0.001 or 0.0003.

Fixed data, model, initialization, seed 17, 50 epochs, batch size 256 and device.

Move the can. What changes?

Shift only the can; pair the source episode and random seed.



Standard



+4 cm in world X

Stable placements

	Standard	Shifted
Learning rate 0.001	●●○○○ 2/5	●●●○○ 3/5
Learning rate 0.0003	●●●●● 5/5	●●●●● 5/5

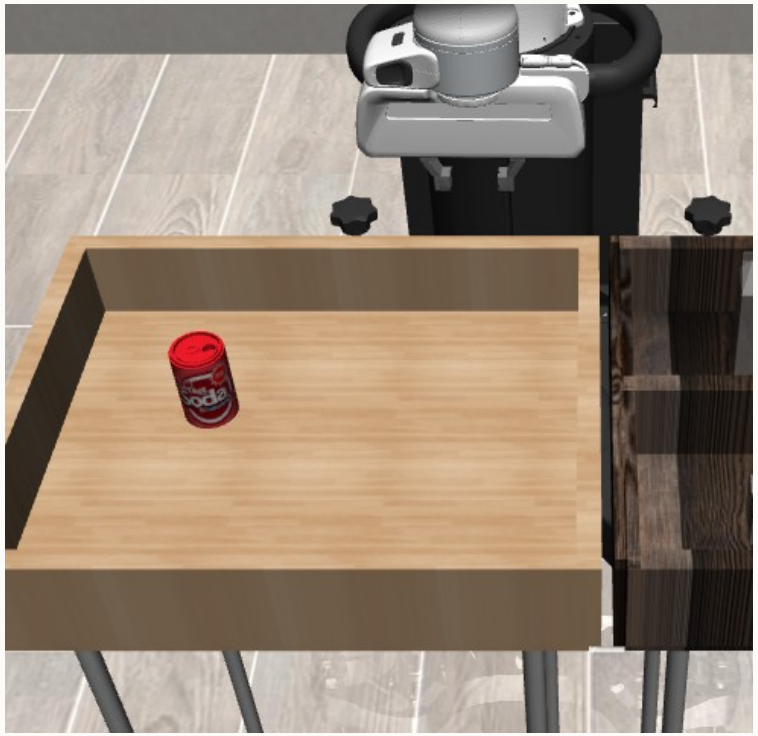
One completed example; five cases are a coarse estimate.

Five matched starts per model

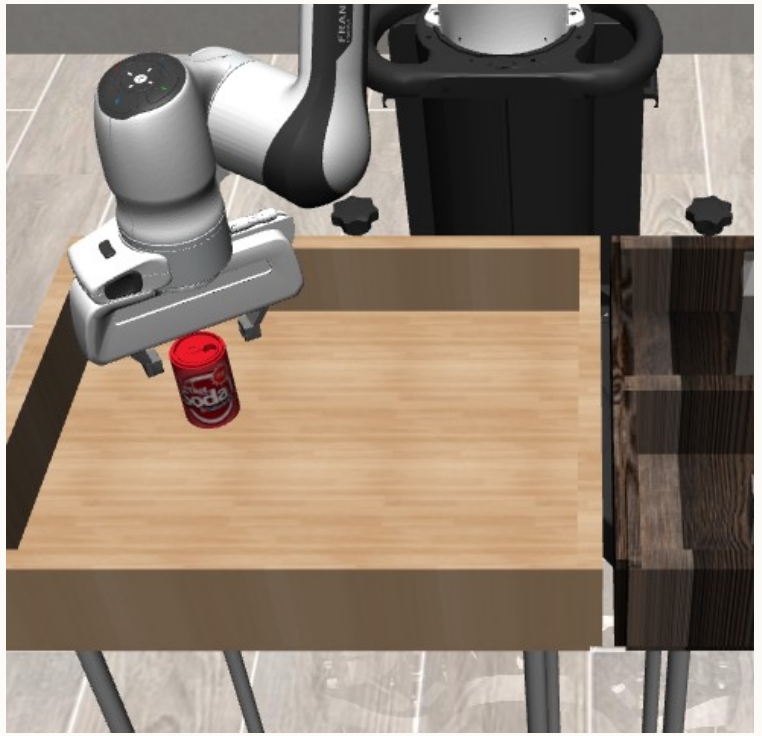
Not a certified out-of-distribution test.

Where does it go wrong?

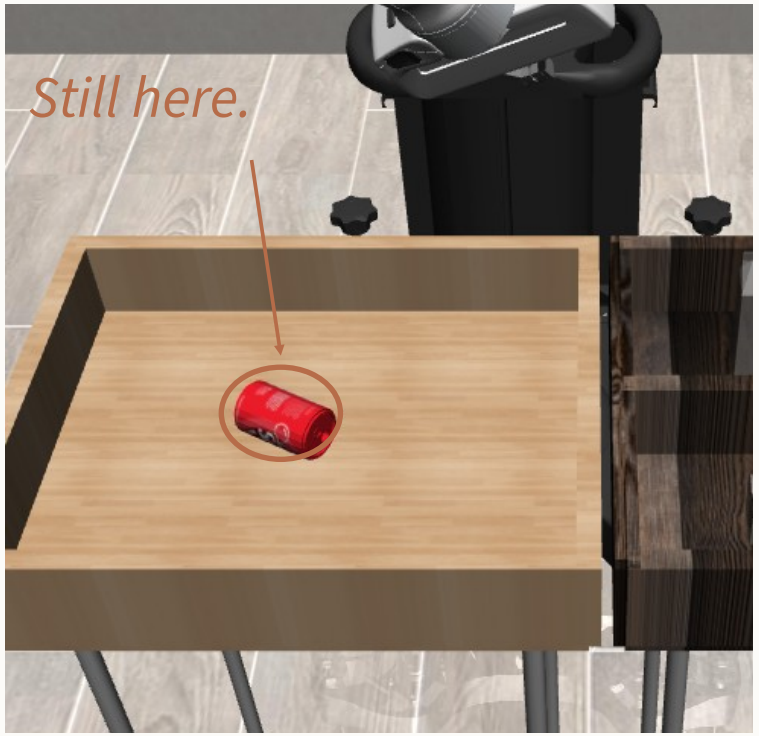
Use frames and state logs to separate the observed error from a possible cause.



Step 1



Step 41



Step 81

Inspect ≥3 steps, mark the first observable error, then propose a test.

What to hand in



PDF

1–2 pages · Q1–Q5



ZIP

Both models · videos · logs

Export the ZIP from Notebook 04. The generated report is an evidence appendix.

Assessment

20	25	30	25
Demonstrations	Training	Evaluation	Failure analysis

A well-explained failure can earn full credit.